

DWT Encryption backup answers.

Use this for rehearsal and Q&A. Do not present it unless asked.

Requirement check

- Group of two
- Journal paper from 2026
- Methodology and findings
- Ten-minute live talk

Paper identity

- ACM TOMM, March 2026
- DOI 10.1145/3769123
- Xu, Gao, Cao, Mou
- Multi-image encryption

DWT reminder

- DWT makes four bands
- LL is low frequency
- Keep LL only
- Image becomes one-quarter size

Chaos reminder

- Same key, same stream
- Tiny change, new stream
- Sequence drives swaps
- Sequence drives XOR

Cube construction

- Compress each image
- Stack LL bands
- Pad smaller images
- Encrypt cube C

Confusion answer

- Changes positions
- Values stay unchanged
- Offsets from chaos
- Nine swap cases

Diffusion answer

- Changes values
- Uses XOR
- Chains previous output
- Creates avalanche effect

PSNR answer

- 32.06 dB average
- Above visual threshold
- Prior work below 28
- Quality stays usable

Security answer

- NIST all passed
- NPCR near ideal
- UACI near ideal
- Entropy almost 8

Critique answer

- Not a formal proof
- Not an AES replacement
- Best for image batches
- Padding wastes space

Q&A anchors

- DWT vs DCT
- Why XOR
- Why cube
- Why padding